

Muhammad Hammad Asif

AI Integration & Python Backend Developer

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Open to full-time & freelance roles — Remote (USA/Canada, UK/Europe) or relocation

Professional Summary

AI Engineer and Python Backend Developer with 5+ years of experience building production-grade AI applications, LLM integrations, and scalable backend systems. Contributed to the Google Gemini fine-tuning pipeline through the Google × Turing collaboration and independently delivered ML solutions on AWS with proven business impact. Skilled in RAG systems, FastAPI, Python, cloud deployment, and full-stack AI product development. Holds an MS in Data Science and a peer-reviewed NLP publication. Available for immediate full-time or freelance engagements worldwide.

Technical Skills

AI / LLM	LangChain, OpenAI API, Gemini API, Hugging Face Transformers, RAG Pipelines, LLM Agents & Function Calling, Prompt Engineering, RLHF, SFT Data Annotation, NLP
ML / Data	TensorFlow, LSTM, SVM, Text Classification, Abstractive Summarization, Pandas, NumPy, Scikit-learn
Backend	Python, FastAPI, Flask, REST API Design, Async Programming, Pydantic, SQLAlchemy, Web Scraping (Playwright, Selenium, BeautifulSoup), PDF Processing
Frontend	React.js, JavaScript (ES6+),
Cloud / DevOps	AWS EC2, Google Cloud Platform (Document AI, OCR), Docker, Ray Cluster Autoscaling, Nginx, CI/CD, Git & GitHub
Databases	PostgreSQL, MySQL, MongoDB, SQLite, Firebase, Supabase, Airtable API

Professional Experience

Toptal Network Member | Full Stack & AI Engineer

February 2026 - Present

Toptal - Remote

- Accepted into Toptal's elite vetted network, a platform that screens out 97% of applicants through rigorous multi-stage technical, problem-solving, and communication assessments.
- Qualified across Full Stack and AI/ML domains, vetted for delivering production-grade Python backends, LLM integrations, and React frontends for international clients.
- Actively available for contract engagements with global teams, bringing 5+ years of end-to-end delivery experience across AI, NLP, and web development.

Python Backend Developer & AI Integration

Jul 2025 - Present

Scanify Dev - Remote

- Integrated OpenAI API features (content generation, image creation, prompt pipeline management) into a live FastAPI backend serving **10–50 active Shopify stores**, automating AI-generated store content and product images at scale.
- Resolved critical production bugs in the AI delivery pipeline, restoring system stability and reducing downtime for paying clients.
- Maintained and extended prompt pipeline architecture to support new product categories and e-commerce use cases.

AI Trainer - Google × Turing Project

Apr 2024 - May 2025

Turing.com - Remote

- Selected for a Google-commissioned project to improve Gemini's reasoning and data analysis capabilities through supervised fine-tuning and expert human evaluation.
- Produced **100–500 high-quality SFT prompt-response pairs** for data analytics tasks, directly contributing to production-grade LLM alignment.
- Performed RLHF annotation ranking and rating Gemini outputs against expert-level analytical standards to align model behavior.

- Designed structured prompt evaluation frameworks for systematic model assessment across diverse scenarios and edge cases.

Python & AI Developer

Jul 2023 - Mar 2024

Recursive Data Ltd. — Multan, Pakistan

- Built a unified parcel tracking platform, scraping real-time delivery data across multiple courier websites, aggregating results via a single tracking ID with Airtable API integration — reducing manual tracking effort for operations teams.
- Integrated ChatGPT into an existing mobile application, enabling intelligent conversational AI features without rebuilding the core system architecture.
- Developed a real-time price comparison tool, simultaneously scraping and aggregating product listings across Amazon, eBay, and multiple e-commerce platforms.

AI & Backend Developer

Jan 2021 - Jun 2023

Zaytrics (Pvt.) Limited — Islamabad, Pakistan

- **TenderML:** Sole developer of an AI document classification system for German construction tender PDFs — processed **thousands of documents**, achieving **95% classification accuracy** using SVM models. Deployed as a FastAPI service on AWS EC2 with Ray cluster autoscaling; served 1–2 enterprise clients independently.
- **GCP InsightDocs:** Independently built a comprehensive document extraction platform using Google Cloud Document AI and OCR, delivered as a client-facing REST API integrated into an existing React frontend.
- **CHappie VOIP SaaS:** Contributed Python backend and API components to a Dutch-origin subscription-based VOIP call centre platform (React, Flask, Node.js, MySQL, MongoDB, Firebase).
- **VOIP Voice Comparison:** Collaborated on a real-time voice comparison feature for VOIP communications using Python, FastAPI, and audio processing, deployed on AWS EC2.
- **SecureTrack:** Built a real-time admin monitoring system for logging user actions, detecting suspicious behaviour, and maintaining audit trails using Node.js and MongoDB.

Selected Projects

Portfolio Scraper & AI Job Score Matcher *Python, FastAPI, LangChain, Playwright, Selenium, SQLite, Docker*

- FastAPI service that scrapes developer portfolios and scores them against job descriptions using LangChain AI analysis, automating candidate-job fit assessment with explainable scoring and Jinja2-rendered reports.

Real-time E-commerce Price Comparison Tool *Python, BeautifulSoup, Selenium, Data Processing, API Integration*

- Comprehensive scraping tool that aggregates and compares product listings across Amazon, eBay, and multiple e-commerce platforms simultaneously, delivering structured real-time pricing output.

FoodScanSaaS Mobile Application *Mobile Development, Barcode Scanning, ChatGPT API, NLP, SaaS Architecture*

- SaaS mobile app that scans food barcodes and uses the ChatGPT API to deliver detailed ingredient analysis, health insights, and natural language explanations to end users.

FlowSketchJS Lightweight Diagramming Library *Pure JavaScript, HTML5 Canvas, SVG, DOM Manipulation*

- Zero-dependency JavaScript library for rendering interactive flow diagrams in-browser with a custom canvas and SVG rendering engine, no external libraries required.

Order Tracking Web Scraping System *Python, Web Scraping, Airtable API, Data Aggregation, Real-time Processing*

- Centralized platform for real-time order tracking across multiple courier and e-commerce platforms using a single tracking ID, with Airtable API integration for structured data storage and retrieval.

Publications

Comparative Evaluation of Text Similarity Metrics for Enhanced Abstractive Summarization on CNN/DailyMail Corpus

Journal of Computational Biology and Informatics (JCBI), 2023

Conducted a comprehensive comparative analysis of semantic, syntactic, and hybrid text similarity metrics and their impact on abstractive summarization quality. Demonstrated measurable improvements in summary coherence and factual accuracy by integrating optimized similarity metrics into the pipeline.

Keywords: Text Similarity Metrics, Abstractive Summarization, NLP, Deep Learning, CNN/DailyMail, ROUGE

Education

MS in Data Science 2020 - 2023

Air University, Islamabad, Pakistan

Research focus: NLP, Text Summarization, Machine Learning Evaluation Metrics. Published in JCBI (2023).

BS in Computer Science 2016 - 2020

Air University, Multan, Pakistan

Final Year Project: Hybrid Abstractive & Extractive Text Summarization System using LSTM and LSA/SVD.

Key Highlights

- **Google × Turing:** Contributed to Gemini AI production training, RLHF annotation, SFT datasets, and prompt evaluation frameworks.
- **Solo production systems:** TenderML (95% accuracy, AWS EC2 + Ray autoscaling) and GCP InsightDocs designed, built, and deployed independently.
- **Live client impact:** LLM features integrated into the Shopify platform serving 10–50 active paying stores.
- **Academic credibility:** MS Data Science, Air University Islamabad + peer-reviewed JCBI publication (2023).
- **Full-stack AI capability:** Python FastAPI/Flask + React.js + AWS/GCP + Docker — end-to-end delivery.
- **Availability:** Immediate start. Open to remote (worldwide) or relocation. Response within 24 hours.